

STOCKHOLMN SKAVSTA, Sweden

WMO: 024853

Lat: 58.789N

Lon: 16.912E

Elev: 43

StdP: 100.81

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-12.9	-18.0	0.8	-15.6	-15.0	1.0	-12.1	12.7	4.9	11.1	4.2	2.2	320	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	26.9	18.6	25.1	17.9	23.2	16.9	20.1	24.2	19.0	23.1	18.1	22.2	4.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	21.9	17.8	12.8	20.9	16.2	11.6	19.8	57.5	24.4	54.0	22.6	51.0	22.3	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.9	7.9	-19.9	29.3	4.2	1.9	-23.0	30.6	-25.4	31.7	-27.8	32.8	-30.9	34.2		
(5)			-20.2	21.4	4.2	1.4	-23.2	22.4	-25.7	23.1	-28.0	23.9	-31.1	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-2.3	-2.0	0.8	5.4	10.3	14.7	17.5	16.7	12.3	7.0	2.5	-0.8		
	DBStd	7.98	4.96	4.79	4.16	3.47	3.54	2.75	2.71	2.75	3.12	3.51	3.82	5.03		
	HDD10.0	1863	383	335	286	143	39	1	0	0	14	104	224	333		
	HDD18.3	4215	641	568	544	387	248	113	49	64	182	352	474	592		
	CDD10.0	731	0	0	0	5	49	142	232	209	83	11	1	0		
	CDD18.3	42	0	0	0	0	0	4	23	15	1	0	0	0		
	CDH23.3	382	0	0	0	1	11	54	207	106	3	0	0	0		
	CDH26.7	53	0	0	0	0	1	9	32	11	0	0	0	0		
(14) Wind	WSAvg	3.7	4.0	4.0	3.9	3.7	3.7	3.7	3.4	3.3	3.5	3.7	3.7	3.9		
(15) Precipitation	PrecAvg	534	40	26	23	30	37	36	65	65	48	49	64	50		
	PrecMax	814	75	57	50	60	70	75	119	216	103	139	129	142		
	PrecMin	377	8	8	0	4	8	4	17	14	20	12	22	7		
	PrecStd	108	18	16	14	15	22	19	30	50	24	33	33	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	8.9	14.1	20.1	24.9	27.9	29.2	28.1	23.9	16.2	12.1	10.9		
		MCWB	7.0	6.5	8.3	12.3	14.9	17.7	19.2	19.4	17.4	13.4	10.2	8.8		
	2%	DB	6.1	6.9	11.1	16.9	22.0	24.8	27.2	26.1	21.1	14.9	10.2	8.1		
		MCWB	5.0	5.2	6.7	10.5	14.0	16.7	18.8	18.8	16.0	12.4	8.9	7.0		
	5%	DB	4.9	5.2	8.9	14.2	19.8	22.9	25.9	24.1	19.2	13.2	9.1	6.9		
		MCWB	4.0	3.8	5.5	8.9	13.3	15.6	18.2	18.2	14.9	11.4	8.3	5.9		
	10%	DB	3.8	4.0	6.8	12.1	17.2	21.0	24.0	22.2	17.8	12.1	8.0	5.2		
		MCWB	3.1	2.9	4.2	7.4	11.7	14.9	17.6	17.3	14.3	10.5	7.3	4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	6.9	9.0	13.1	16.7	19.2	21.6	21.8	18.5	14.2	11.0	9.1		
		MCDB	7.7	7.8	13.4	19.0	21.7	24.5	25.6	25.4	21.7	15.7	12.2	10.4		
	2%	WB	5.2	5.2	7.0	10.8	15.0	17.6	20.3	20.2	16.7	12.9	9.3	7.0		
		MCDB	5.7	6.1	10.2	15.6	19.7	22.8	24.7	23.9	20.0	14.3	10.1	7.6		
	5%	WB	3.8	4.0	5.6	9.5	13.6	16.5	19.3	19.0	15.6	12.0	8.2	5.7		
		MCDB	4.4	4.9	8.1	13.9	18.6	21.2	23.8	22.4	18.4	13.4	8.9	6.2		
	10%	WB	2.8	2.9	4.3	8.0	12.4	15.5	18.4	18.1	14.7	11.0	7.1	4.5		
		MCDB	3.4	3.6	6.0	11.0	16.8	19.9	22.8	21.6	17.2	12.2	7.5	5.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	6.0	7.7	9.8	10.5	10.0	10.0	9.8	8.9	6.7	4.8	4.9		
		MCDBR	5.4	6.4	10.7	14.0	14.1	13.2	13.0	12.0	10.9	7.6	6.4	5.2		
	5% WB	MCWBR	4.9	5.2	7.1	8.6	7.7	6.8	5.9	6.0	6.5	5.8	5.8	4.9		
		MCDWR	4.9	5.7	9.9	12.8	12.4	11.4	10.8	10.4	9.7	6.8	5.7	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.321	0.338	0.375	0.372	0.361	0.389	0.380	0.347	0.337	0.310	0.270		
		taud	2.245	2.305	2.364	2.325	2.404	2.462	2.405	2.426	2.498	2.426	2.283	2.172		
	Ebn at Noon		539	699	803	829	860	875	838	818	793	678	497	415		
		Edh at Noon	45	69	87	107	105	101	105	96	76	62	44	34		
(44) All-Sky Solar Radiation	RadAvg		0.33	0.92	2.42	3.97	5.25	5.80	5.31	4.22	2.81	1.26	0.42	0.23		
	RadStd		0.04	0.15	0.25	0.39	0.43	0.28	0.49	0.39	0.33	0.19	0.09	0.03		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46)	Station Only	+0.51	N/A	N/A	N/A	N/A	N/A	-161	-183	N/A
(47)	Regional (53 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	-122	-149	N/A

Nomenclature: See separate page